

2.2 Integrating using Trigonometric Identities

Review:

In this section we will be making use of some trig. Identities. In particular, you should recall:

$$\sin^2(x) + \cos^2(x) = 1 \quad \rightarrow \quad \tan^2(x) + 1 = \sec^2(x) \quad \& \quad 1 + \cot^2(x) = \csc^2(x)$$

$$\cos(2x) = 1 - 2\sin^2(x) \Leftrightarrow \sin^2(x) = \frac{1}{2}[1 - \cos(2x)]$$

$$\cos(2x) = 2\cos^2(x) - 1 \Leftrightarrow \cos^2(x) = \frac{1}{2}[1 + \cos(2x)]$$

$$\sin(2x) = 2\sin(x)\cos(x)$$

General Idea:

How could you use one of these identities to find $\int \cos^3(x) \, dx$?

One possible answer is to use the fact that $\cos^2(x) = 1 - \sin^2(x)$ in order to rewrite the integral as:

$$\int (1 - \sin^2(x)) \cos(x) \, dx.$$

Then, perform a u -substitution by letting $u = \sin(x)$.

This implies $du = \cos(x)dx$, and the integral becomes:

$$\int (1 - u^2) du = u - \frac{1}{3}u^3 + C.$$

We conclude that $\int \cos^3(x) \, dx = \sin(x) - \frac{1}{3}\sin^3(x) + C$.

Note: There are examples where a “nice” substitution is not possible.

Example 1: Calculate the exact value of $\int_0^{\pi/4} \cos^2(x) \, dx$.

For this problem we use a double angle identity. Specifically, we calculate:

$$\begin{aligned} \int_0^{\pi/4} \cos^2(x) \, dx &= \int_0^{\pi/4} \frac{1}{2}(1 + \cos(2x)) \, dx = \left(\frac{1}{2}x + \frac{1}{4}\sin(2x) \right) \bigg|_0^{\pi/4} \\ &= \left(\frac{1}{2}\left(\frac{\pi}{4}\right) + \frac{1}{4}\sin\left(\frac{2\pi}{4}\right) \right) - \left(0 + \frac{1}{4}\sin(0) \right) = \frac{\pi}{8} + \frac{1}{4}. \end{aligned}$$

Example 2: Find $\int \sin^4(x) dx$.

This problem requires us to use a double angle identity twice. We calculate:

$$\begin{aligned}\int \sin^4(x) dx &= \int (\sin^2(x))^2 dx = \int \left(\frac{1}{2}(1 - \cos(2x)) \right)^2 dx \\&= \frac{1}{4} \int (1 - 2\cos(2x) + \cos^2(2x)) dx = \frac{1}{4} \int \left(1 - 2\cos(2x) + \frac{1}{2}(1 + \cos(4x)) \right) dx \\&= \frac{1}{4} \int \left(\frac{3}{2} - 2\cos(2x) + \frac{1}{2}\cos(4x) \right) dx = \frac{1}{4} \left(\frac{3}{2}x - \sin(2x) + \frac{1}{8}\sin(4x) \right) + C \\&= \frac{3}{8}x - \frac{1}{4}\sin(2x) + \frac{1}{32}\sin(4x) + C.\end{aligned}$$

Strategy for Evaluating $\int \sin^m(x)\cos^n(x)dx$:

1. If the power of cosine is odd, save a cosine factor and use the identity $\cos^2(x) = 1 - \sin^2(x)$ to write the remaining factors in terms of sine. Then, make the substitution: $u = \sin(x)$.
2. If the power of sine is odd, save a sine factor and use the identity $\sin^2(x) = 1 - \cos^2(x)$ to write the remaining factors in terms of cosine. Then, make the substitution: $u = \cos(x)$.
3. If the powers of both sine and cosine are even use the double angle identities.

Note: If both m and n are odd numbers one can use 1 or 2.

Example 3 (you try): Find $\int \sin^3(x)\cos^5(x) dx$

We note that: $\int \sin^3(x)\cos^5(x) dx = \int \sin^2(x)\cos^5(x)\sin(x) dx = \int (1 - \cos^2(x))\cos^5(x)\sin(x) dx$.

We let $u = \cos(x)$ from which we obtain: $du = -\sin(x)dx \Rightarrow -du = \sin(x)dx$. So,

$$\begin{aligned}\int (1 - \cos^2(x))\cos^5(x)\sin(x)dx \\&= \int -(1 - u^2)u^5 du = \int (u^7 - u^5) du = \frac{1}{8}u^8 - \frac{1}{6}u^6 + C = \frac{1}{8}\cos^8(x) - \frac{1}{6}\cos^6(x) + C.\end{aligned}$$

Note: In the above example there are solutions which look different from the above solutions which are equivalent to the above solutions and hence correct. This is a reoccurring theme when it comes to trigonometric integration. For example, if one decided to save a factor of cosine (rather than sine) in (a) he or she would obtain an answer with sine's in it rather than cosines (which would still be correct).

Example 4: Find $\int \tan^4(x) \sec^4(x) dx$

In this case we want to make use of the identity: $\sec^2(x) = 1 + \tan^2(x)$. We do so in the following way:

$$\int \tan^4(x) \sec^4(x) dx = \int \tan^4(x) \sec^2(x) \sec^2(x) dx = \int \tan^4(x) (1 + \tan^2(x)) \sec^2(x) dx.$$

Now, we make the substitution $u = \tan(x)$ which implies $du = \sec^2(x) dx$. Now we calculate:

$$\begin{aligned} \int \tan^4(x) (1 + \tan^2(x)) \sec^2(x) dx &= \int u^4 (1 + u^2) du \\ &= \frac{1}{5} u^5 + \frac{1}{7} u^7 + C = \frac{1}{5} \tan^5(x) + \frac{1}{7} \tan^7(x) + C. \end{aligned}$$

Example 5 (you try): Find $\int \tan^5(\theta) \sec^7(\theta) d\theta$

Solution: We write the integral as:

$\int \tan^4(\theta) \sec^6(\theta) \sec(\theta) \tan(\theta) d\theta = \int (\sec^2(\theta) - 1)^2 \sec^6(\theta) \sec(\theta) \tan(\theta) d\theta$. Now, we let $u = \sec(\theta)$ which implies $du = \sec(\theta) \tan(\theta) d\theta$. So, we have that

$$\begin{aligned} \int (\sec^2(\theta) - 1)^2 \sec^6(\theta) \sec(\theta) \tan(\theta) d\theta &= \int (u^2 - 1)^2 u^6 du = \int (u^4 - 2u^2 + 1) u^6 du \\ &= \int (u^{10} - 2u^8 + u^6) du = \frac{1}{11} u^{11} - \frac{2}{9} u^9 + \frac{1}{7} u^7 + C = \frac{1}{11} \sec^{11}(\theta) - \frac{2}{9} \sec^9(\theta) + \frac{1}{7} \sec^7(\theta) + C. \end{aligned}$$

Strategy for Evaluating $\int \tan^m(x) \sec^n(x) dx$:

1. If the power of secant is even, save a factor of $\sec^2(x)$ and use $\sec^2(x) = 1 + \tan^2(x)$ to express the remaining factors in terms of $\tan(x)$. Then, make the substitution $u = \tan(x)$.
2. If the power of tangent is odd and there is at least one secant factor, save a factor of $\sec(x) \tan(x)$ and use $\tan^2(x) = \sec^2(x) - 1$ to express the remaining factors in terms of $\sec(x)$.

Note: Strategies similar to these also work when the integral contains cosecants and cotangents.

Note: When we cannot apply strategy 1 or 2, we need to use ingenuity to find integrals which contain secants and cosecants.

Useful formulas:

You should recall that

$$\int \tan(x) dx = -\ln|\cos(x)| + C = \ln|\sec(x)| + C$$

Now, consider $\int \sec(x) dx = \int \left(\frac{\sec(x)(\sec(x) + \tan(x))}{\sec(x) + \tan(x)} \right) dx$. If we make the substitution:

$u = \sec(x) + \tan(x)$, then $du = \sec^2(x) + \tan(x)\sec(x) dx$ and the integral on the right becomes:

$$\int \frac{1}{u} du = \ln|u| + C = \ln|\sec(x) + \tan(x)| + C.$$

Now, we have formulas for the integral of secant and tangent: $\int \tan(x) dx = \ln|\sec(x)| + C$ and

$\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$. These will be useful.

$$\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$$

Example 6: Find $\int \sec^3(x) dx$.

We will integrate by parts. We let $u = \sec(x)$ and $dv = \sec^2(x)dx$. These two equations imply: $du = \sec(x)\tan(x)dx$ and $v = \tan(x)$. Now, if we apply the integration by parts formula we obtain:

$$\int \sec^3(x) dx = \sec(x)\tan(x) - \int \tan^2(x)\sec(x) dx.$$

Now, we turn our attention to $\int \tan^2(x)\sec(x) dx$. We notice that:

$\int \tan^2(x)\sec(x) dx = \int (\sec^2(x) - 1)\sec(x) dx = \int \sec^3(x) dx - \int \sec(x) dx$. Now we plug this result into the equation for the integral of secant to the third, and we obtain:

$$\int \sec^3(x) dx = \sec(x)\tan(x) - \left(\int \sec^3(x) dx - \int \sec(x) dx \right)$$

$$\Leftrightarrow 2 \int \sec^3(x) dx = \sec(x)\tan(x) + \int \sec(x) dx$$

$$\Leftrightarrow 2 \int \sec^3(x) dx = \sec(x)\tan(x) + \ln|\sec(x) + \tan(x)| + C$$

$$\Leftrightarrow \int \sec^3(x) dx = \frac{1}{2}(\sec(x)\tan(x) + \ln|\sec(x) + \tan(x)|) + C'.$$

Note: This example shows that integration by parts can be quite useful when trying to find a difficult indefinite integral involving secants and/or tangents.